

Panel Data Econometrics

Chapter 1 – Panel Data Structure Opportunities and Challenges

Introduction

The Moodle course will contain many resources.

- Slides
- Additional references
- Additional exercises
- Exams from the previous year

If you have any question, feel free to contact me:

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Introduction

In this first chapter, we discuss what is panel data structure, what advantages it brings and the challenges associated to their statistical treatment.

Part 1 – Reminders regarding Econometrics

Part 2 – What is panel data structure

Part 3 – Illustrations

Part 4 – Decomposing the variability

Part 5 – Softwares and resources

Reminders regarding Econometrics

The linear regression model

$$y = X\beta + \varepsilon$$

$$y = \begin{pmatrix} y_1 \\ \dots \\ y_n \end{pmatrix}$$

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \dots \\ \varepsilon_n \end{pmatrix}$$

$$\beta = \begin{pmatrix} \beta_0 \\ \dots \\ \beta_K \end{pmatrix}$$

$$X = \begin{pmatrix} 1 & x_{11} & \dots & x_{1K} \\ \dots & \dots & x_{ik} & \dots \\ 1 & x_{n1} & \dots & x_{nK} \end{pmatrix}$$

Reminders regarding Econometrics

Hypothèses

A1. Linearity

$$y_i = x_{i1}\beta_1 + x_{i2}\beta_2 + \cdots + x_{iK}\beta_K + \varepsilon_i$$

A2. Full Rank

There is no exact linear relationship between exogeneous variables.

A3. Exogeneity of independent variables

$$E[\varepsilon_i | x_{i1}, x_{i2}, \dots, x_{iK}] = 0$$

A4. Homoskedasticity and non-autocorrelated error terms

$$\text{var}[\varepsilon_i | X] = \sigma^2 \quad ; \quad \text{cov}[\varepsilon_i, \varepsilon_j | X] = 0$$

A5. Exogenous generation of data

The process which generates data in X is independent to the process generating ε .

A6. Normal distribution of the error terms.

$$\varepsilon_i \sim N(0, \sigma^2)$$

Reminders regarding Econometrics

Removing hypothesis A4 : $E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|\mathbf{X}] = \sigma^2\boldsymbol{\Omega}$

The OLS estimator is not biased, is consistent, but is not efficient anymore.

The GLS estimator is BLUE, but cannot be estimated if $\boldsymbol{\Omega}$ is not known.

The FGLS is a nice solution if you are ready to make appropriate hypothèses.

The OLS estimator can still be used, but the variance-covariance matrix has to be correctly estimated.

$$Var[\hat{\boldsymbol{\beta}}|\mathbf{X}] = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\boldsymbol{\Omega}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1} \neq \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

Reminders regarding Econometrics

Removing hypothesis A3 : $E[\varepsilon_i|\mathbf{X}] = \eta_i$

The OLS estimator will be biased, and not consistent.

The instrumental variable estimator is consistent, provided that we have identified appropriate instruments to estimate it.

Instruments are related to the problematic variable(s), but not directly related to the explained variable.

The Hausman-Wu test can be used to compare instrumental variable and OLS estimators, so that we can identify the presence of endogeneity.

What is panel data structure

Data is said to have a panel data structure if it combines the cross-sectional and the time-series dimensions.

Cross-sectional dimension : You have information regarding several entities or individuals.

Time-series dimension : Observations are collected over multiple time-periods.

What is panel data structure

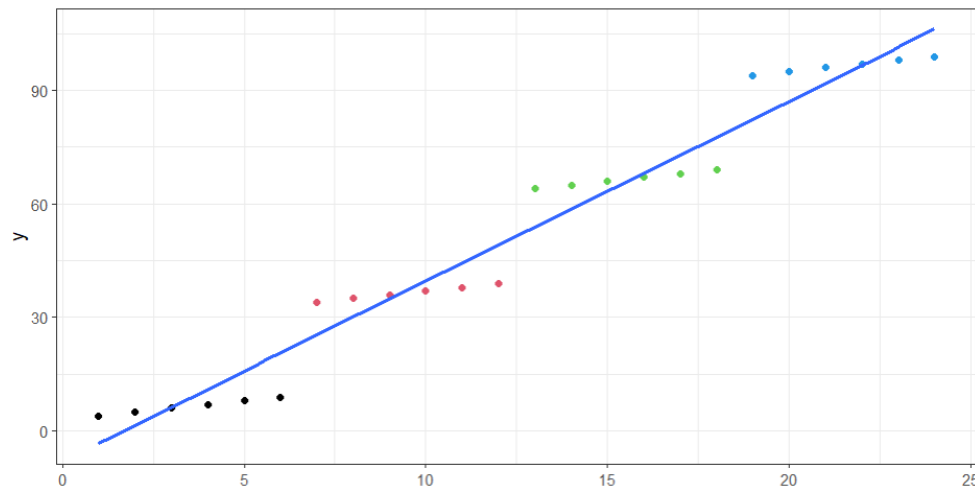
Working with panel data has several advantages:

- Richness of information and larger datasets.
 - $N \times T$ observations !
- Control over individual and time-specific heterogeneity.
- Limits the issue of multicollinearity.
- One can study the dynamics of individual behavior.
- Makes it easier to deal with endogeneity.

What is panel data structure

Working with panel data demands some care:

- Model specification is particularly important.
 - Misspecifying the model could lead to important biases in the estimations



- Panel data are quite often messy.
 - Extreme values
 - Missing values (unbalanced panel)

Illustration

Let's work on an example :

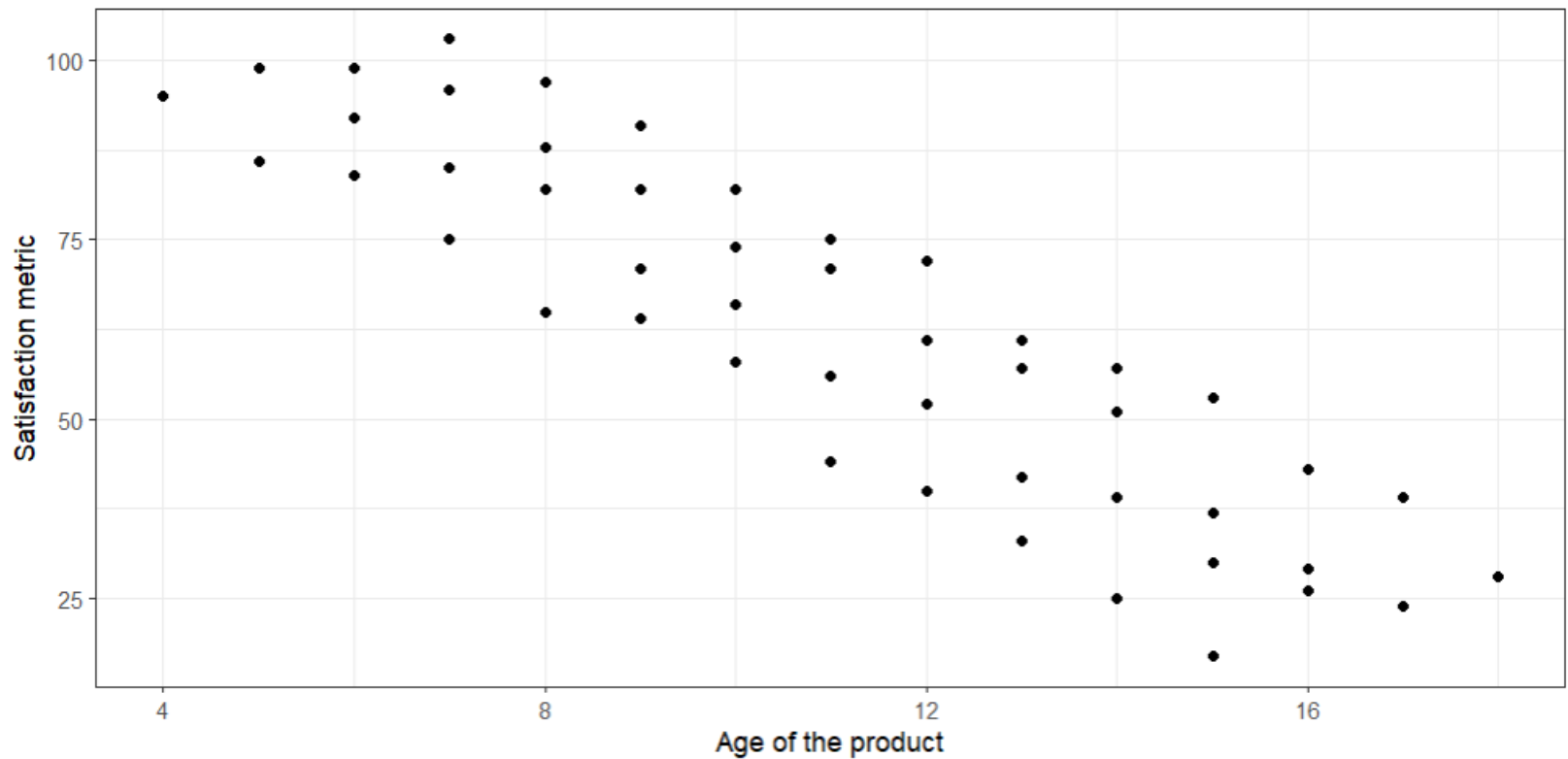
Dataset : Productsatisfaction (you can find it on Moodle)

- Product : Product ID.
- Product_age : Time passed since the commercialization.
- Satisfaction : recorded satisfaction from a panel of users.

```
> head(data)
# A tibble: 6 × 4
  product period product_age satisfaction
  <dbl>   <dbl>      <dbl>      <dbl>
1       1       1         4          95
2       1       2         5          99
3       1       3         6          99
4       1       4         7         103
5       2       1         5          86
6       2       2         6          92
```

Illustration

Graphical representation with ggplot()



Illustration

OLS estimation agrees with the plot : the older a product, the lesser the satisfaction.

```
> reg = lm(satisfaction ~ product_age, data=data)
> summary(reg)
```

call:

```
lm(formula = satisfaction ~ product_age, data = data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-21.7521	-8.6263	-0.5393	8.9172	16.9188

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	127.4942	5.1024	24.99	<2e-16 ***
product_age	-5.9161	0.4405	-13.43	<2e-16 ***

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

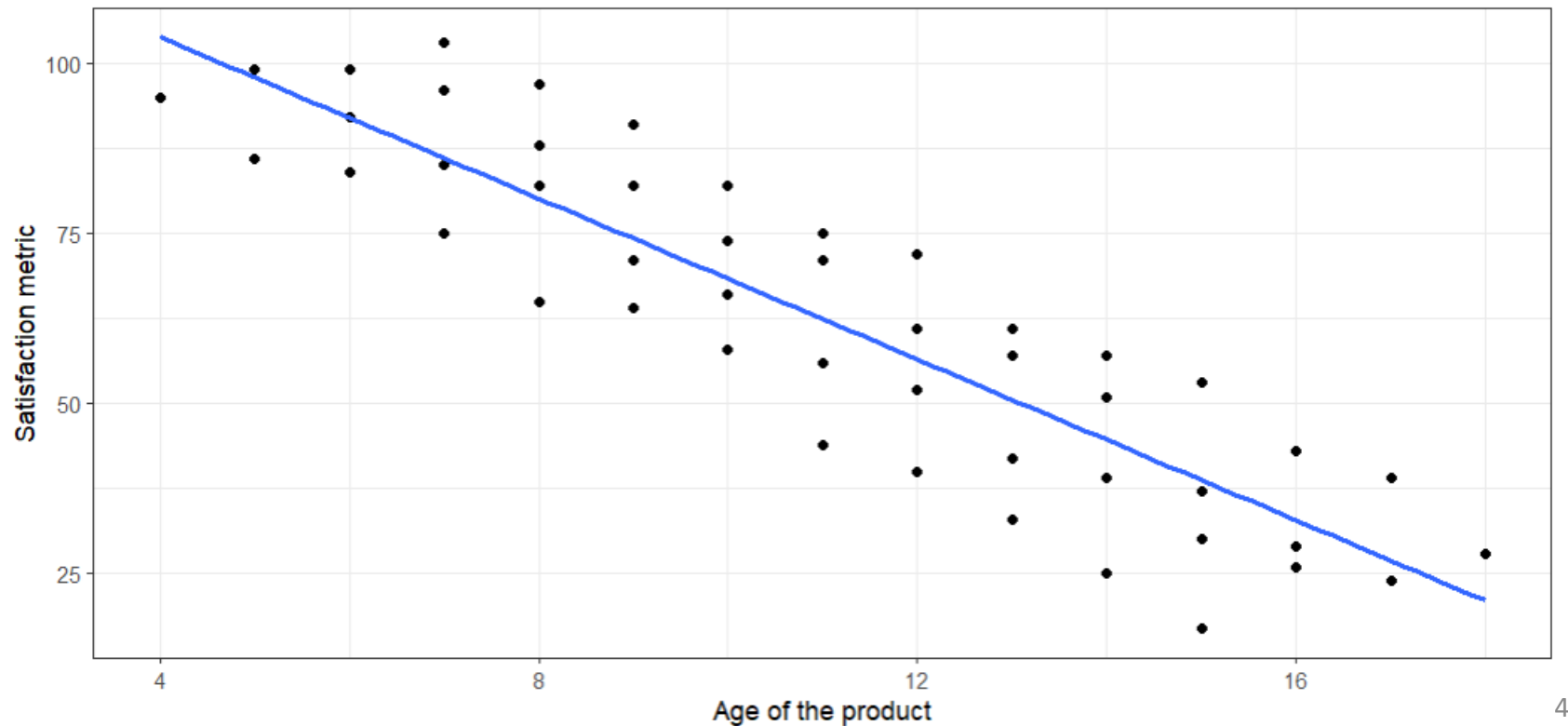
Residual standard error: 11.07 on 46 degrees of freedom

Multiple R-squared: 0.7968, Adjusted R-squared: 0.7924

F-statistic: 180.4 on 1 and 46 DF, p-value: < 2.2e-16

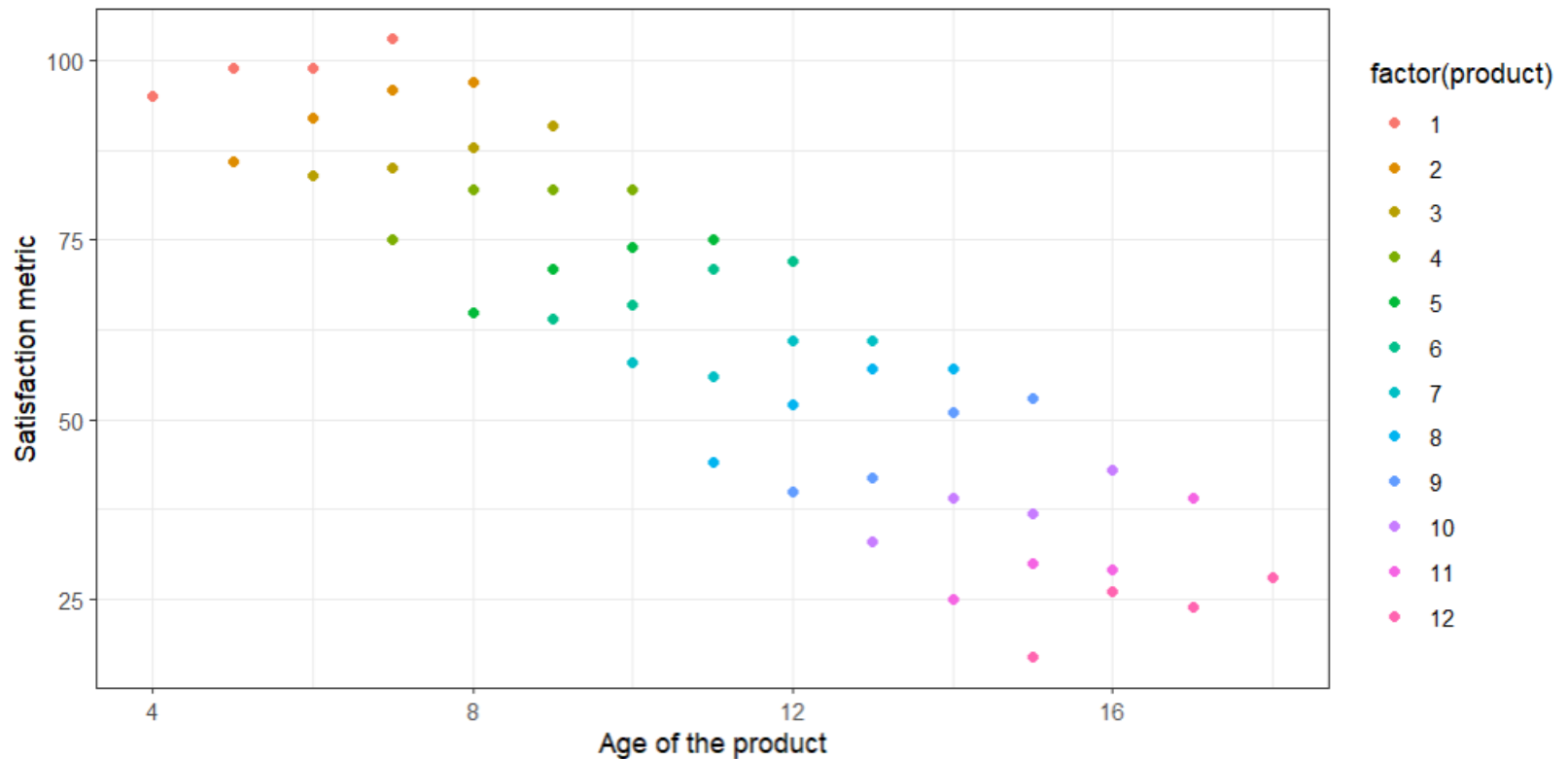
Illustration

Graphical representation with `ggplot()` and `geom_smooth()`



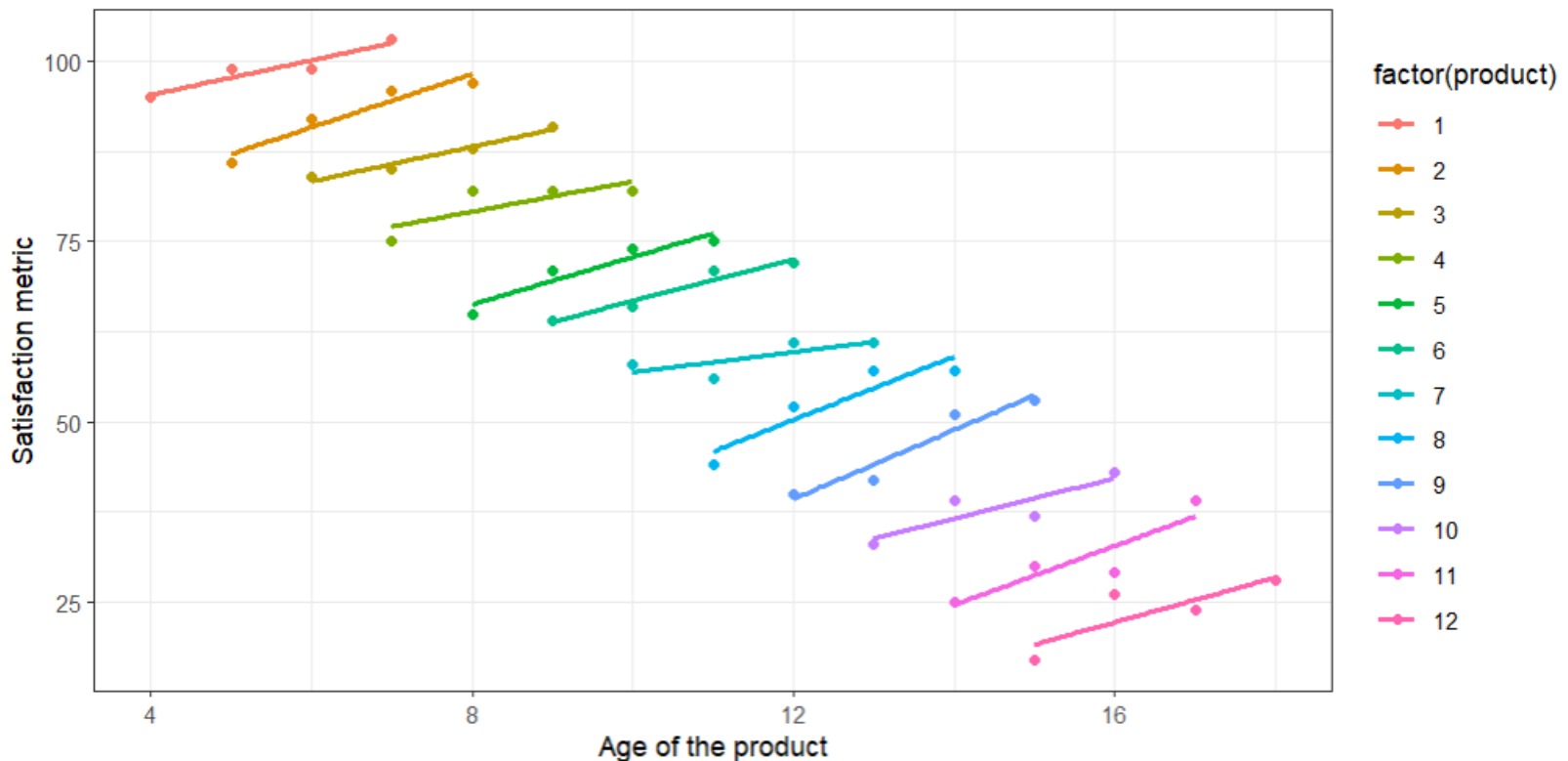
Illustration

There is however some hidden information in the data !



Illustration

There is however some hidden information in the data !
Are our previous conclusions really relevant?



Decomposing the variability

Let's consider our variable of interest:

$$y_{it}, i = 1, \dots, N, t = 1, \dots, T$$

$$\mathbf{y} = \begin{pmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_N \end{pmatrix} \text{ with } \mathbf{y}_i = \begin{pmatrix} y_{i1} \\ y_{i2} \\ \vdots \\ y_{iT} \end{pmatrix} \text{ for } i = 1, \dots, N$$

Decomposing the variability

Several decompositions are available when it comes to the variability of a variable.

Total variability = inter-indiv. variability + intra-indiv. variability

Total variability = inter-temp. variability + intra-temp. variability

Total variability = inter-indiv. variability + inter-temp. variability + inter-indiv. variability
+ intra-indiv.-temp variability

The estimators we use in panel data econometrics mostly rely on those decompositions.

Decomposing the variability

Let's display the mathematical notations.

Global average

$$y_{..} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T y_{it}$$

Cross-sectional average

$$y_{i.} = \frac{1}{T} \sum_{t=1}^T y_{it}$$

Time average

$$y_{.t} = \frac{1}{N} \sum_{n=1}^N y_{it}$$

Decomposing the variability

Total variability

$$Var_{tot} = \sum_{i=1}^N \sum_{t=1}^T (y_{it} - y_{..})^2$$

Inter-indiv. variability

$$Var_{inter-ind.} = T \sum_{i=1}^N (y_{i.} - y_{..})^2$$

→ Captures the systematic differences between individuals

Inter-temp. variability

$$Var_{inter-temp.} = N \sum_{t=1}^T (y_{.t} - y_{..})^2$$

→ Captures the systematic differences between periods

Decomposing the variability

Intra-indiv. variability

$$Var_{intra-ind.} = \sum_{i=1}^N \sum_{t=1}^T (y_{it} - y_{i.})^2$$

→ Captures the differences between the situation of an individual with their average over all periods

Intra-temp. variability

$$Var_{intra-temp.} = \sum_{t=1}^T \sum_{i=1}^N (y_{it} - y_{.t})^2$$

→ Captures the differences between the situation of an individual with the average of the period.

Decomposing the variability

Intra-indiv.-temp. variability

$$Var_{intra-ind.-temp.} = \sum_{i=1}^N \sum_{t=1}^T (y_{it} - y_{i.} - y_{.t} + y_{..})^2$$

Decomposing the variability

Coming back to our example :

```
> data %>%  
+   summarise('total variation' = sum((satisfaction-mean(satisfaction))^2),  
+             'inter-ind.' = sum((indmean-mean(satisfaction))^2),  
+             'intra-ind.' = sum((satisfaction-indmean)^2)  
+   )  
# A tibble: 1 × 3  
  `total variation` `inter-ind.` `intra-ind.`  
    <dbl>         <dbl>         <dbl>  
1    27762.         27001.           760.
```

It looks like variations in our data are mainly due to inter-individual variations and not to intra-individual variations.

One can wonder in this setting if the age of a given product really matters.

Decomposing the variability

Using matrix annotation, we introduce the **between operator**.

Note : we focus on the individual dimension

$$\underbrace{B}_{(NT,NT)} = \left(I_N \otimes \frac{J_T}{T} \right) \text{ so that } By = \begin{pmatrix} \frac{J_T}{T} y_1 \\ \frac{J_T}{T} y_2 \\ \dots \\ \frac{J_T}{T} y_N \end{pmatrix}$$

By is therefore a vector containing the individual average, repeated T times for each individual i .

Decomposing the variability

Using matrix annotation, we introduce the **within operator**.

Note : we focus on the individual dimension

$$\underset{(NT,NT)}{\underbrace{W}} = I_N \otimes \left(I_T - \frac{J_T}{T} \right) \text{ so that } Wy = \begin{pmatrix} \left(I_T - \frac{J_T}{T} \right) y_1 \\ \left(I_T - \frac{J_T}{T} \right) y_2 \\ \dots \\ \left(I_T - \frac{J_T}{T} \right) y_N \end{pmatrix}$$

Wy is therefore a vector containing the distances to the individual average for each period t and each individual i .

Softwares and resources

Using R, you can deal with panel data using the plm package.

<https://cran.r-project.org/web/packages/plm/plm.pdf>

The package has been developed and maintained by Yves Croissant, who also wrote a book on the topic :

<https://onlinelibrary.wiley.com/doi/book/10.1002/9781119504641>

Softwares and resources

Using Stata, panel models are used with the xt* commands.

xtset : to declare your panel datastructure

xtreg : to run a panel data linear model

... and others : xtprobit, xtlogit ...

<https://www.stata.com/features/panel-longitudinal-data/>

Softwares and resources

Using SAS, you may want to use the procedure PROC PANEL

<https://support.sas.com/resources/papers/proceedings19/3584-2019.pdf>

Softwares and resources

Using Python, the standard panel data models are available in the library linearmodels :

For the fixed-effect model :

from linearmodels import PanelOLS

For the random-effect model :

from linearmodels import RandomEffects

<https://pypi.org/project/linearmodels/>